

Sahil Satish Kumar

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Education

NOVA School of Science and Technology

Lisbon, Portugal

MSc in Computer Science and Engineering

September 2022 – March 2025

- GPA: **18/20 (A)** on the ECTS grading system). Thesis grade: **18/20**.
- Relevant coursework: Machine Learning, Deep Learning, Visualization and Data Analytics, High Performance Computing, Internet Application Design and Implementation, Cloud Computing Systems, Human-Computer Interaction.

Lady Margaret Hall, Oxford University

Oxford, United Kingdom

Summer School in Artificial Intelligence and Machine Learning

August 2022

- Oxford Grade: **85/100** (Letter Grade Equivalent: **A+**).
- Implemented Machine Learning algorithms and Deep Learning models in PyTorch, focused on computer vision.

NOVA School of Science and Technology

Lisbon, Portugal

BSc in Computer Science and Engineering

September 2019 – July 2022

- GPA: **18/20 (A)** on the ECTS grading system).
- Scientific Talents in Artificial Intelligence Scholarship, Calouste Gulbenkian Foundation.
- NOVA Young Talent Award 2020-21 (Best student of the first year of the course).
- Relevant coursework: Artificial Intelligence, Probability and Statistics, Databases, Software Engineering, Distributed Systems, Algorithms and Data Structures.

Experience

Sky Portugal

Lisbon, Portugal

Apprentice Backend Developer

July 2025 – August 2025

- **Built, published, and deployed a reusable React component library** (TypeScript+Vite) across **3** services, improving UI consistency, development speed, and maintainability via full **CI/CD pipeline** while working in **Agile**.
- **Enhanced** and **added client retention service endpoints** with **Java, Spring AI, and Ollama**, generating **AI-powered error summaries** and **configuration suggestions**, improving user comprehension, reducing support requests, and delivering improvements iteratively in an **Agile workflow**.

Project AI Fights Space Debris, Neuraspace

Lisbon, Portugal

Scholarship Holder (Master Thesis)

March 2024 – March 2025

- **Increased satellite collision forecasting accuracy to 80%** (10x above baseline) by **engineering and training RNN/CNN-based Spatio-Temporal Graph Neural Networks** on transformed, evenly spaced TLE datasets.
- **Built** a discrete-time dynamic **space conjunction graph** using large-scale orbital data for improved debris risk prediction.
- **Published and presented** a research paper at the **9th Causal Inference Workshop, UAI 2024**.

Human-Computer Interaction Institute, Carnegie Mellon University

Pittsburgh, PA, USA

Visiting Researcher

October 2024 – December 2024

- **Developed an LLM-powered pipeline for Human-Centered Explainable AI (XAI)**, converting GNN explanation subgraphs into **clear, human-readable narratives** to enhance interpretability and trust.
- **Designed a user study** to evaluate generated explanation clarity, trust, and decision-support effectiveness.

NOVA LINC3, NOVA School of Science and Technology

Lisbon, Portugal

Undergraduate Researcher

January 2022 – July 2022

- **Achieved 81% predictive accuracy** and **4x improvement over baselines** by **developing** a novel causal explanation method for GNN-based doctor referral recommendations.
- **Processed** and **modelled** sensitive healthcare datasets to construct a **doctor referral graph** for explainable AI research.

Skills

Programming Languages: Java, Python, C/C++, Kotlin, JavaScript, TypeScript.

Machine Learning & Data Science: PyTorch, Tensorflow, Scikit-Learn, Pandas, NumPy, Tableau, Spark, Spring AI.

Software Development: HTML5, CSS3, Bootstrap, React, Redux, Flutter, Express.js, Spring Boot, Spring AI.

Databases: MySQL, MongoDB, Redis, Neo4j.

DevOps: CI/CD, Agile, Jenkins, Git, Microsoft Azure, Kubernetes, Docker.

Other: Ollama, CUDA, MPI, Linux, Bash, MS Office 365.

Languages

Portuguese (Native), **Gujarati** (Native), **Hindi** (Native), **English** (Professional(C1)).